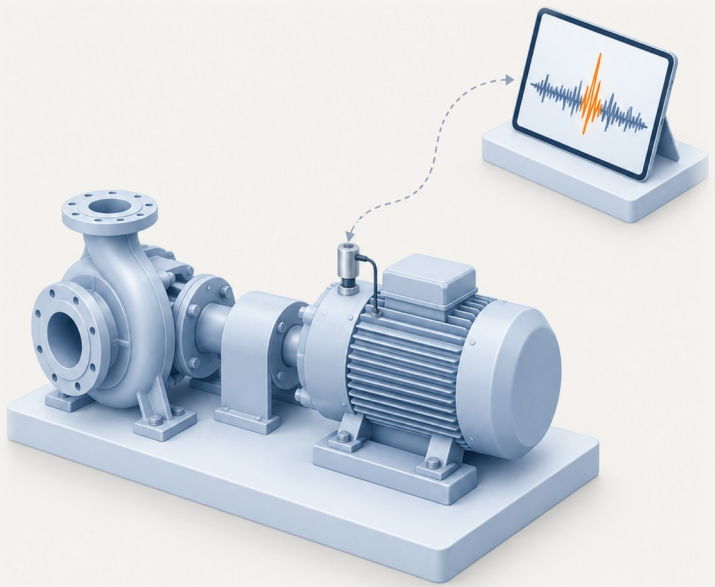
NUS-ISS · PATTERN RECOGNITION SYSTEMS

MACHINESENSE

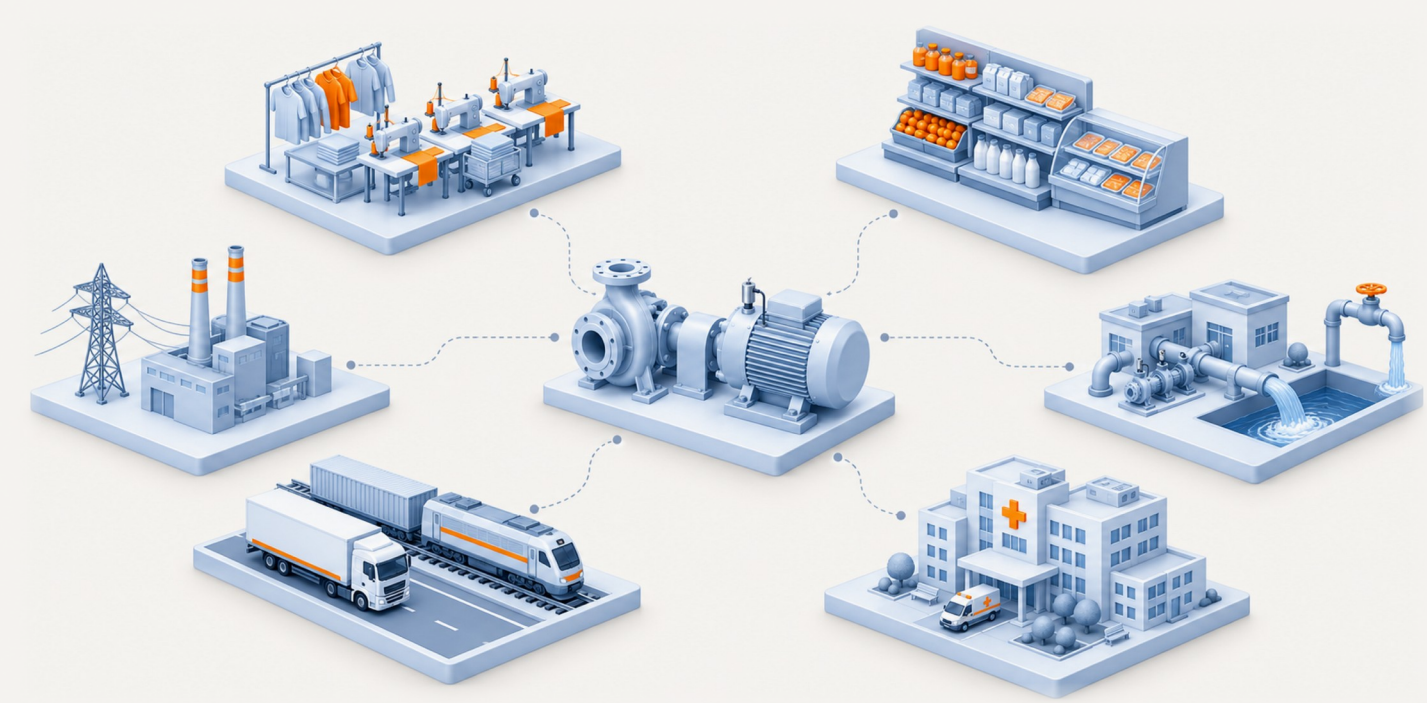
Multi-sensor fault diagnosis for **rotating machinery**, from a five-second reading to an evidence-supported inspection decision.

Bhuvesh Kumar · A0243823H
MTech AI Systems · Practice Module



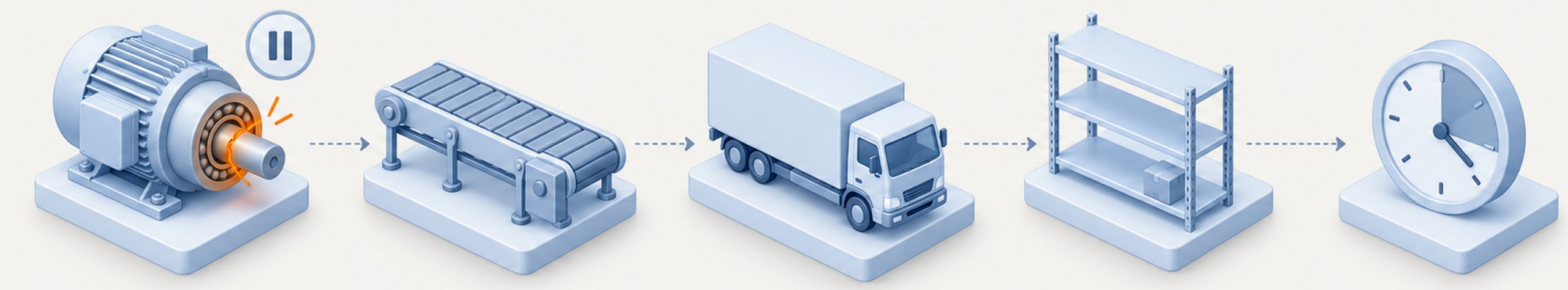
MOTIVATION

Machines keep our lives moving



The clothes we wear, the food on the shelf, the power in the socket, the water in the tap, everything that moves and every hospital that stays open all run on rotating machinery.

What happens when a machine breaks down without warning?



One bearing gives way and the line stops. Production halts, deliveries are late, shelves empty, and the repair costs far more than it would have before the failure.

The problem: three questions from one recording

Q1

Is it healthy?

Only 49 healthy recordings exist, all from one folder, and healthy vibration changes with shaft speed.

Q2

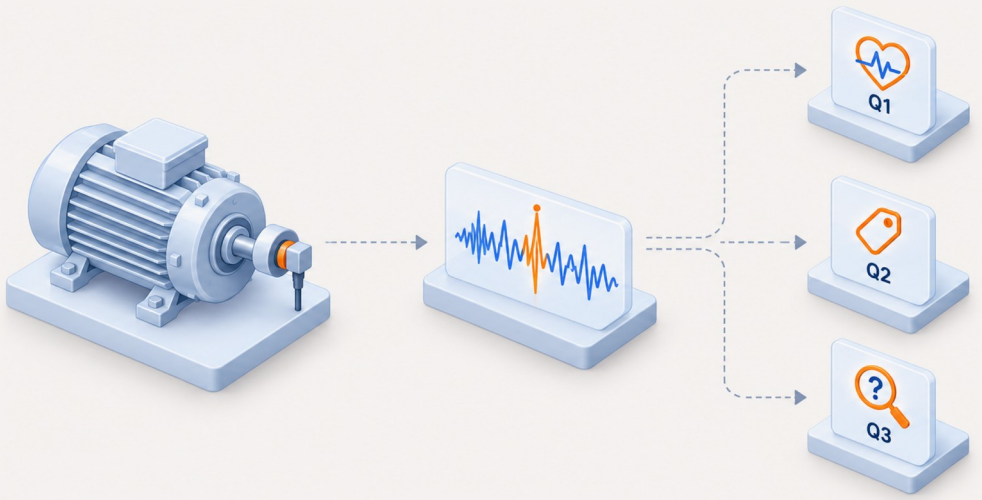
Which known fault is it?

Published scores are often inflated: windows of one recording end up in both training and testing.

Q3

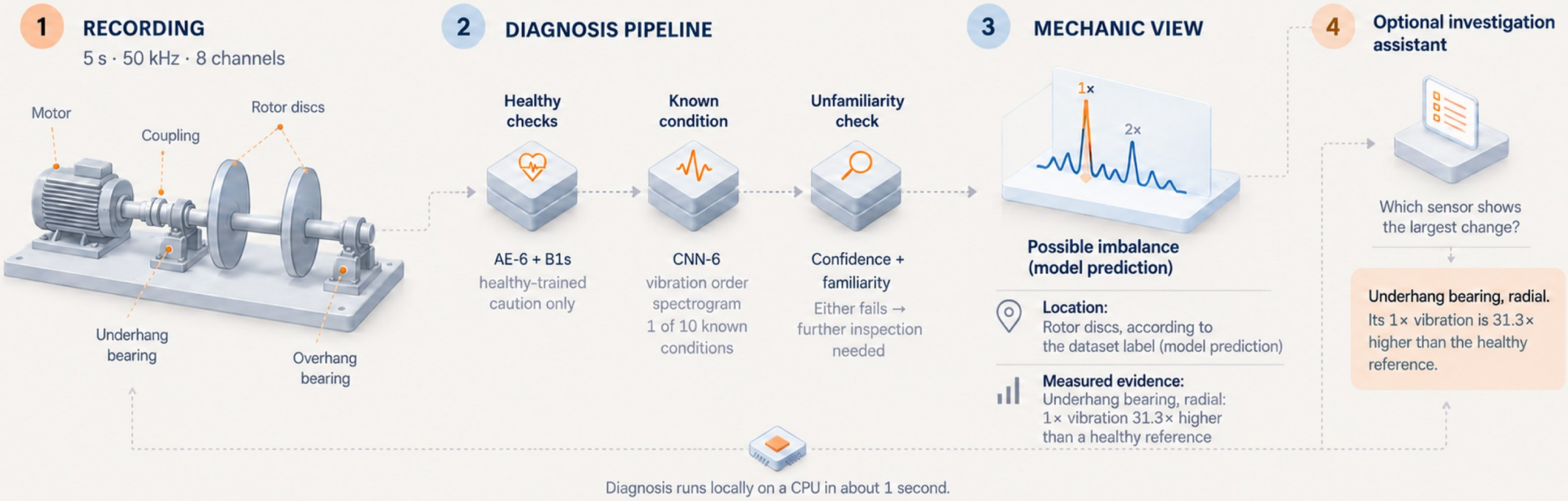
Is it a fault never seen before?

A closed-set classifier must pick a known label, so it can name an unseen fault wrongly, and confidently.

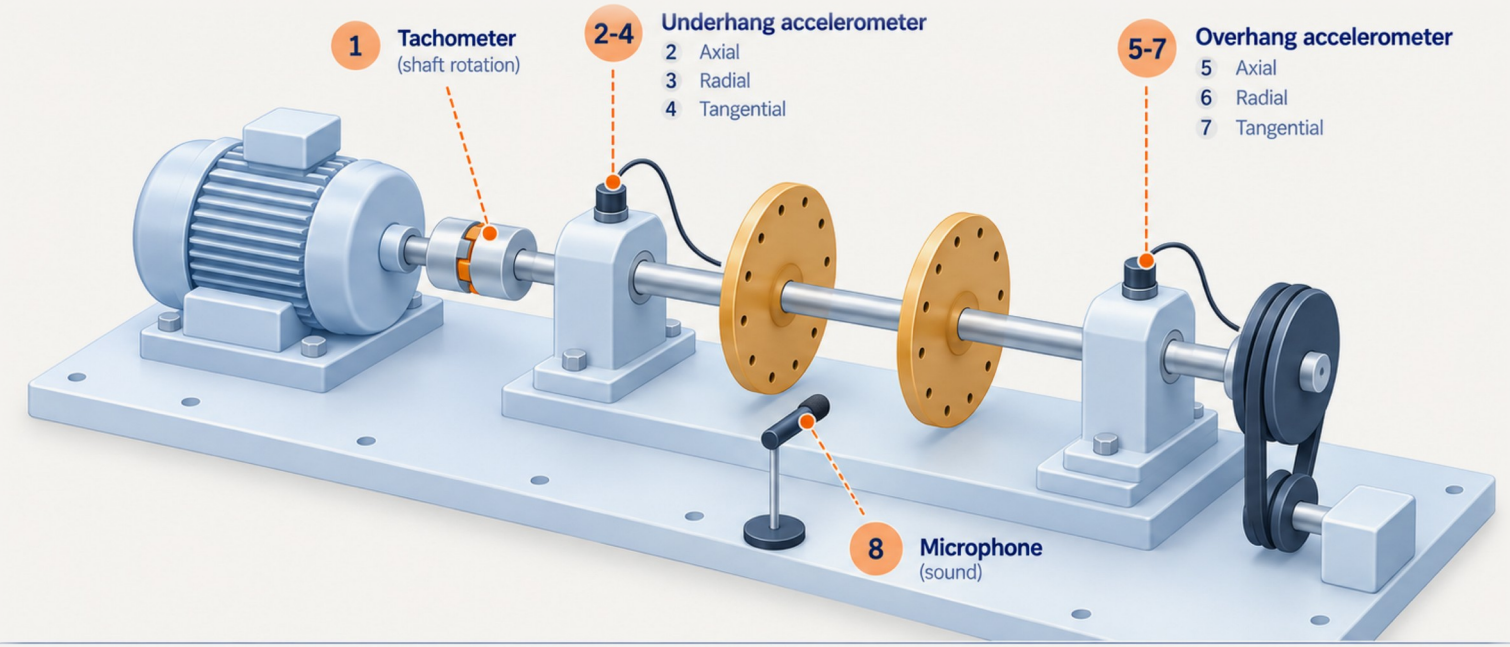


And a mechanic needs the evidence and the uncertainty behind each answer, not just a label.

Introducing MachineSense



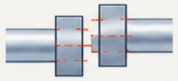
The data: one test rig, 1,951 recordings



Typical fault locations



Imbalance
masses added to rotor discs



Horizontal / vertical misalignment
shaft offset at coupling



Bearing faults
ball, cage or outer race, in either housing

1,951 recordings · 5 s each · 50 kHz · 8 channels

1,951
recordings, 5 s each at 50 kHz

8
channels: tachometer, 2 × 3-axis accelerometers, microphone

10
conditions: normal and 9 induced faults

49
healthy recordings only, all in one folder (333 imbalance)

Source: MAFAULDA, Federal University of Rio de Janeiro. Faults are deliberately induced on one bench-top rig.

Data pipeline, part A: prepare the dataset once

A. Prepare the dataset (once)

1

Raw archive

1,951 CSVs in 6 folders
5 s • 250,000 rows × 8 channels • 50 kHz
No index or label file.

2

Index + convert

Folder path → 10-class label + operating setting
Filename → nominal speed
CSV → float32 NumPy
metadata.csv

3

Audit

Tachometer glitches: 228 files
Microphone bursts: 168 files
22 kHz line • saturation • DC offsets
Flags and cleaning rules; no recording removed.

4

Split before windows

804 development recordings
3 folder-grouped folds
Closed, speed-block and severity tests sealed
Split hash logged.



Data pipeline, part B: recording to model inputs

B. Prepare each recording

5

Measure shaft speed

Filename speed is ~2.3% high.
Cross-check tachometer edges, tachometer FFT and accelerometer harmonic comb.
Uploads estimated from the signal; filename not required.



6

Clean whole recording

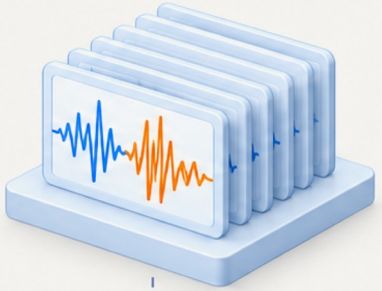
Interpolate microphone bursts.
Band-stop 21.870–22.195 kHz.
Low-pass at 10 kHz.
Tachometer untouched.



7

Cut windows

9 windows, 1 s each, 0.5 s hop.
Remove DC in channels 1–7 per window.

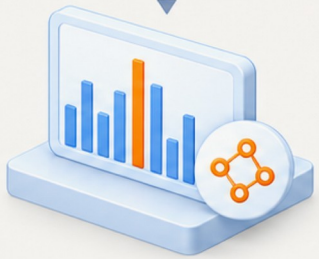


C. Represent each window

8a **Physics features**

142 per window: RMS, peak, crest, kurtosis, shaft orders, axial/radial ratios, bearing-envelope lines, microphone bands.

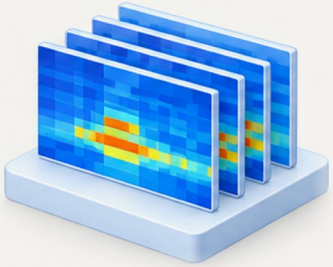
Logistic regression + B1s.



8b **Order spectrogram**

Shaft-angle resampling.
6 vibration channels × 128 shaft-order bins (0.5×–150×) × 25 time frames.

CNN-6 + AE-6.

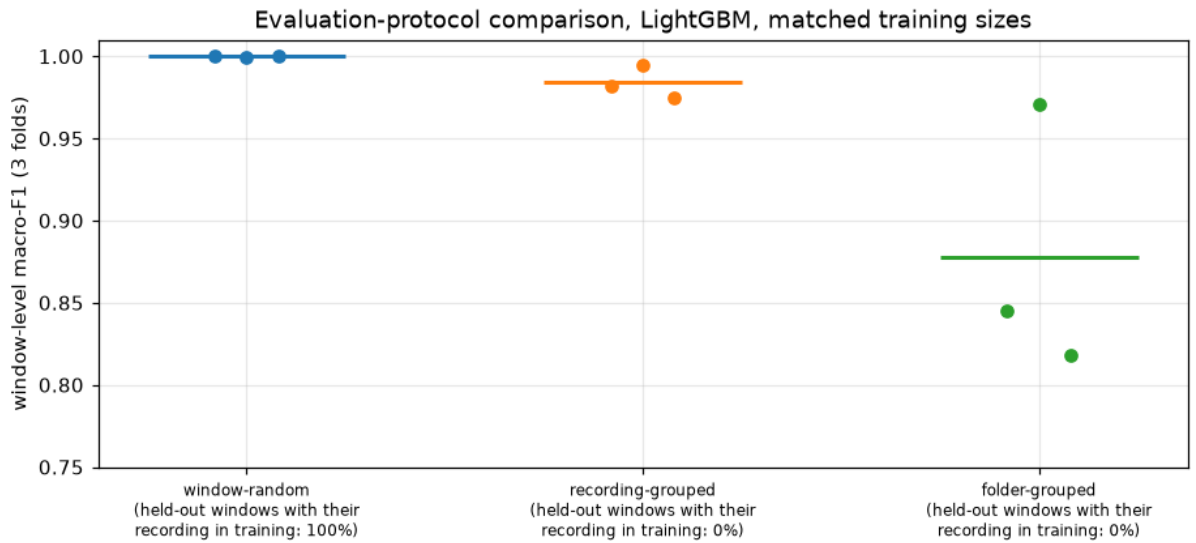


Normalisation statistics from training windows only.

How the data is split



Keeping the evaluation honest



1.000 → 0.984 → 0.878

Same LightGBM, same training size. Only what is held out changes: random windows, whole recordings, whole source folders. Real output of notebook 05b, development data.

1 **Split by recording, before windowing**

Windows of one recording never cross the train and test line.

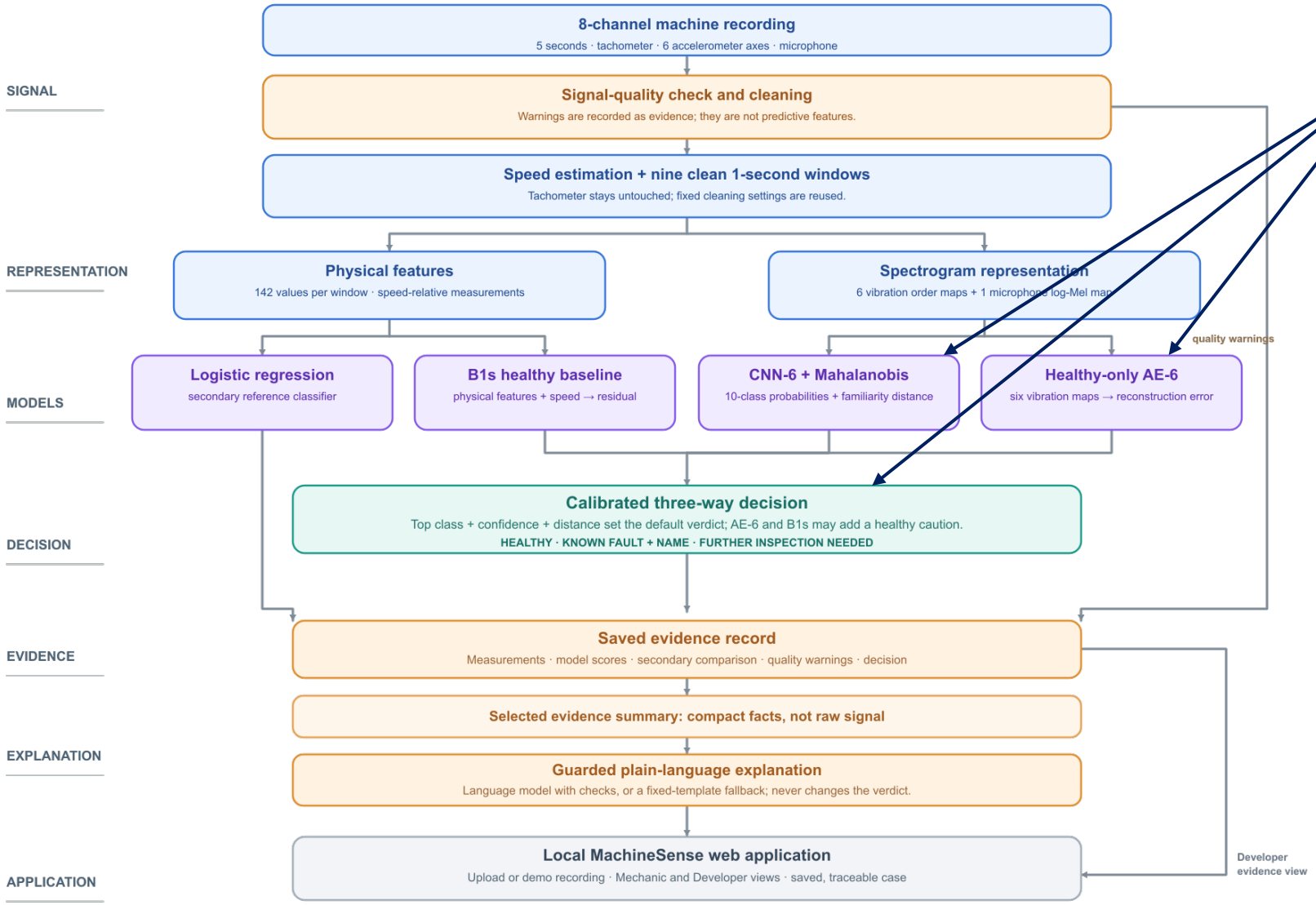
2 **Folder-grouped development folds**

804 development recordings in 3 folds; whole source folders are held out.

3 **Sealed tests, opened once**

Closed split 290, speed block 776, severity 370. One final run after models, thresholds and rule were frozen.

System architecture

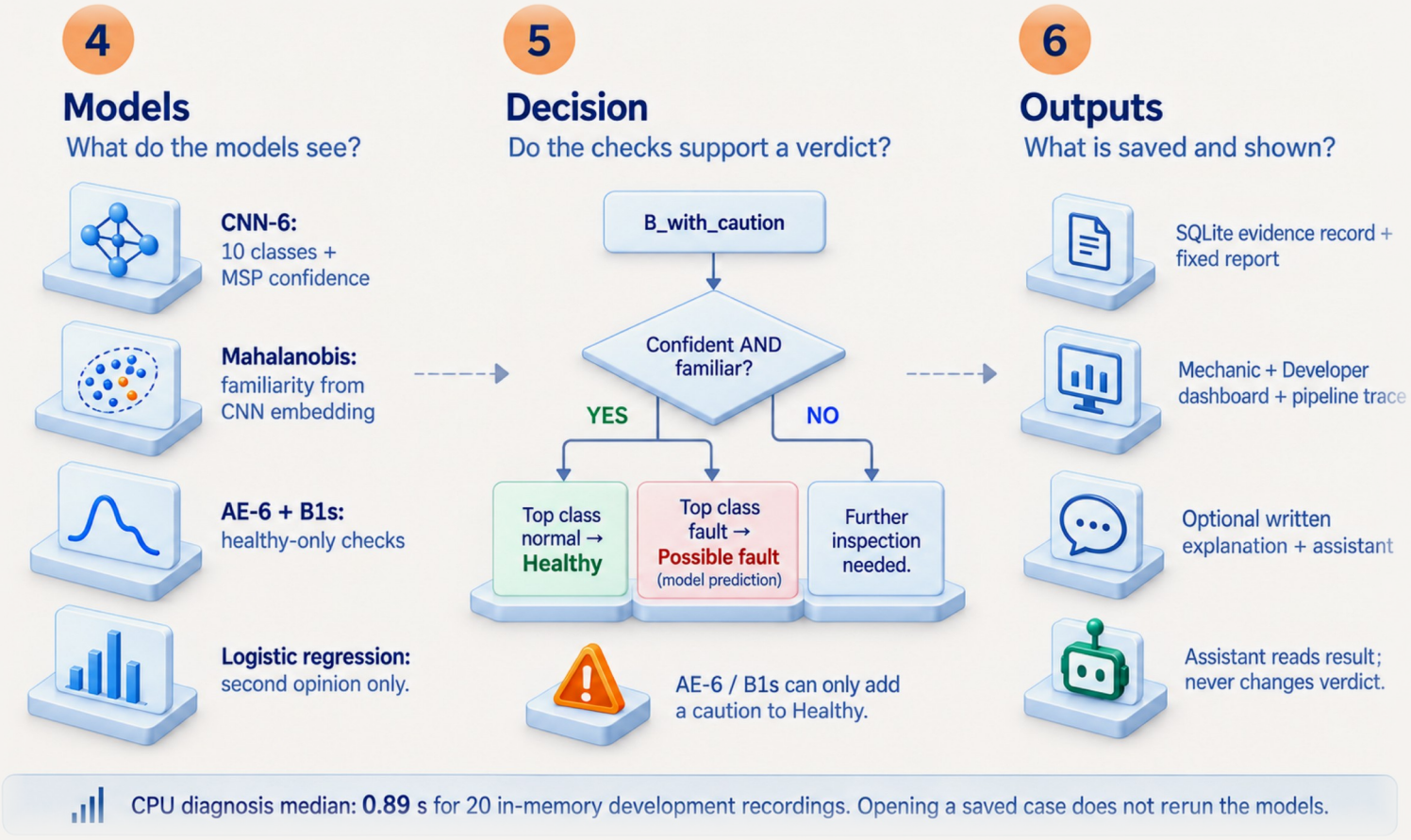


Experimentations to improve the eval metrics and design an appropriate system

Diagnosis, part 1: from recording to features



Diagnosis, part 2: models, decision and outputs



Q1 Is it healthy?

AE-6 + B1s, trained on healthy data only. They can add a caution, never overturn a verdict.

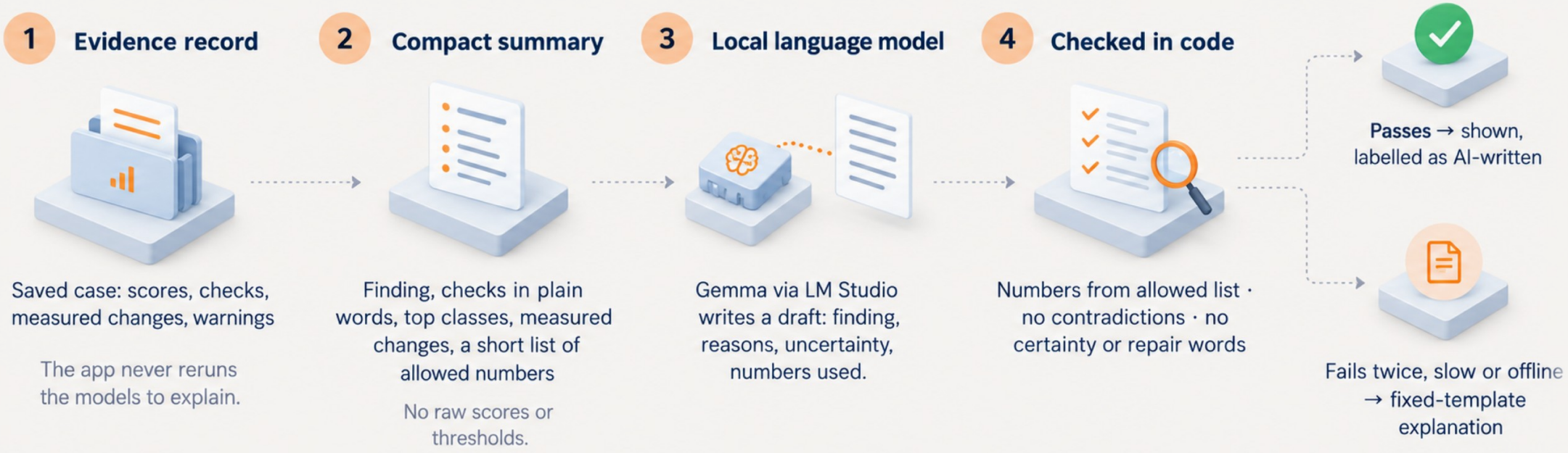
Q2 Which known fault?

CNN-6 on the order spectrogram; logistic regression as a second opinion.

Q3 Is it unfamiliar?

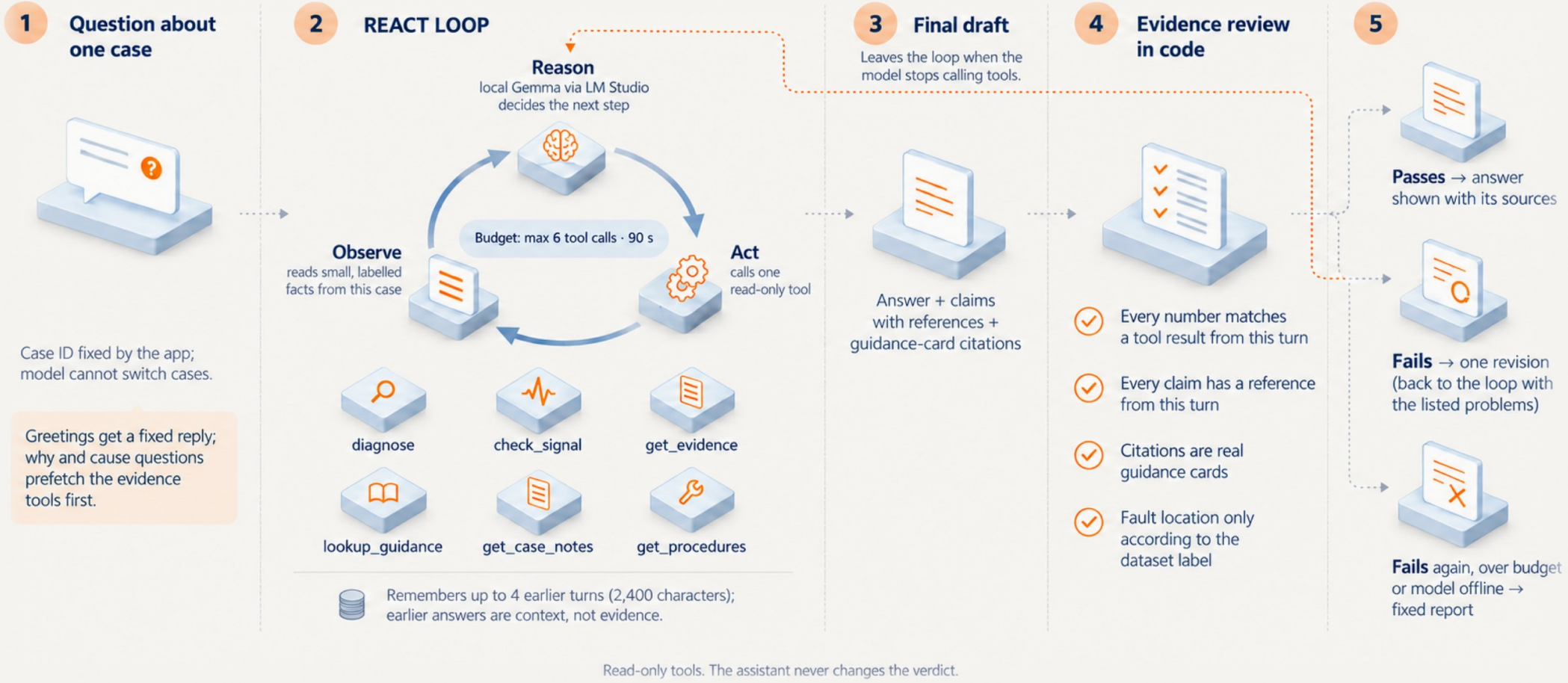
MSP confidence + Mahalanobis familiarity. If either fails: further inspection.

How explanations stay honest



Prompt version, model, latency and fallback reason are logged with every explanation.

The assistant: a ReAct agent with guardrails



Why these models

input

Order spectrogram

A fault stays at the same shaft order when speed changes. A plain spectrogram could not separate 0.5× from 1× at 12 Hz.

0.989

CNN-6

Mean development macro-F1 over 3 seeds. ResNet-18: 0.969 and about 10× slower on a CPU. CNN-7 with microphone: 0.982 and a shortcut risk.

healthy only

AE-6 + B1s

Need no fault examples, so the 49 against 333 imbalance does not matter. Similar AUROC, but they fail differently.

≈ 0.875

MSP + Mahalanobis

Highest deployable macro AUROC for unfamiliar faults. One shared covariance stays stable with few recordings per class.

frozen

Fixed decision rule

Easy to inspect and explain, frozen before the test. A learned combiner was worse on both criteria.

≈ 4 ms

Small on purpose

CNN-6 has 0.6 M parameters and runs in about 4 ms per recording on a CPU, so it is cheap to retrain 81 times.

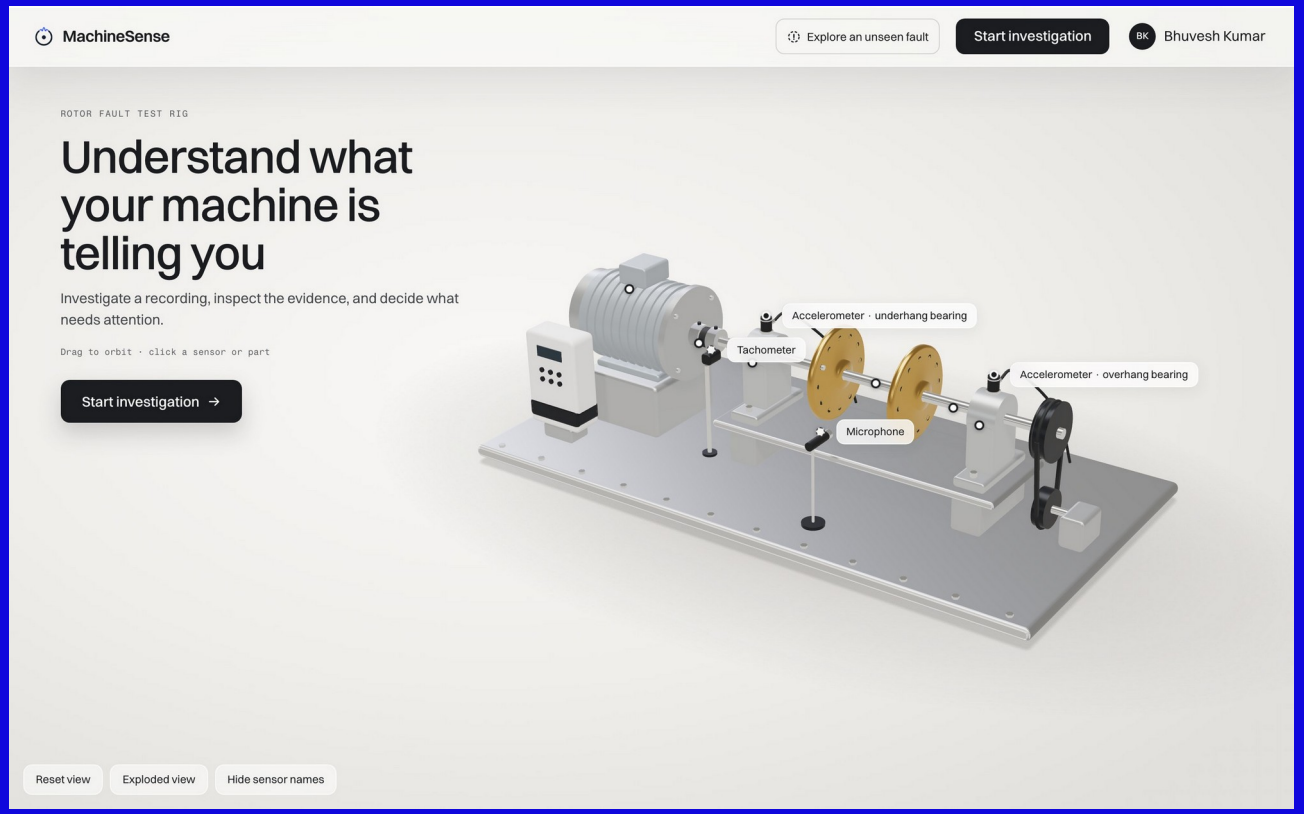
Tech stack: everything runs locally

Data and training	Python 3.12 · PyTorch on Apple MPS · scikit-learn · LightGBM · MLflow tracking
Inference	CPU only · Hashed model bundle · 885 ms median diagnosis
Backend	FastAPI · Pydantic v2 · SQLite case store
Frontend	React 18 · Vite · Tailwind · three.js 3D rig
Assistant	Gemma via LM Studio · No data leaves the machine · 6 read-only tools · Answers checked in code

Live demo

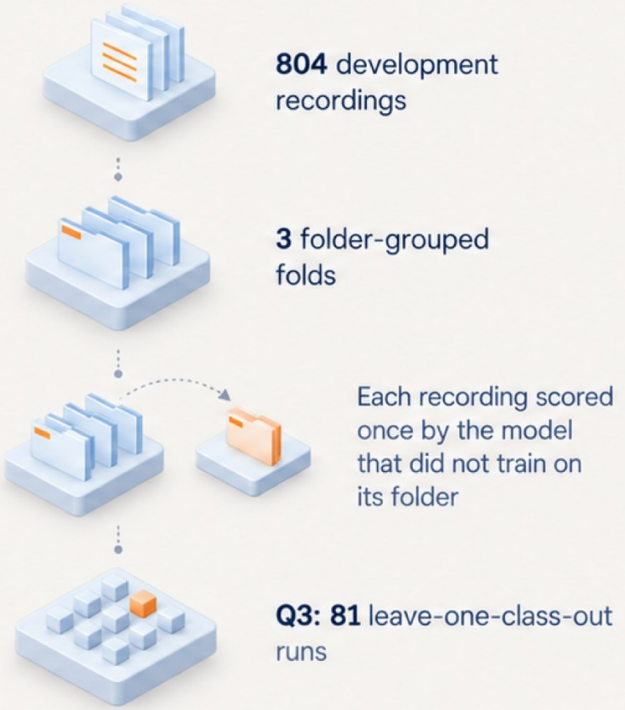
github.com/bhuveshsharma09/MachineSense

Bhuvesh Kumar · NUS-ISS



What data each evaluation uses

1 Development (model choice and research)



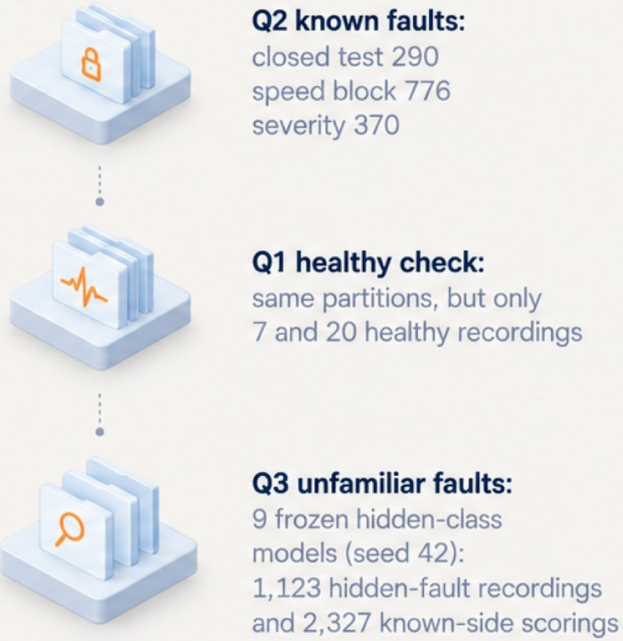
804 development recordings

3 folder-grouped folds

Each recording scored once by the model that did not train on its folder

Q3: 81 leave-one-class-out runs

2 Final evaluation (once, frozen system)



Q2 known faults:
closed test 290
speed block 776
severity 370

Q1 healthy check:
same partitions, but only
7 and 20 healthy recordings

Q3 unfamiliar faults:
9 frozen hidden-class models (seed 42):
1,123 hidden-fault recordings and 2,327 known-side scorings

3 Assistant (development cases)



D7 pilot:
15 cases × 5 questions × 3 arms = 225 answers, scored blind to arm

App-9:
6 scenarios × 6 questions × 2 modes = 72 answers

Only the middle column uses sealed data. Models, thresholds and rule were frozen first.

Final evaluation: sealed tests, opened once

Frozen system, one run on 28 September 2026, one test rig. Reserved recordings only.

Q1 Is it healthy?

0

faults called healthy

- **283 / 283** faults detected on the closed split
- **5 of 7** healthy recordings sent to inspection
- **19 of 20** on the unseen-speed block

Healthy calibration covered only 18 to 38 Hz.

Q2 Which known fault?

0.916

ten-class macro-F1, 290 recordings

- **0.845 to 0.972** 95 % interval
- **0.901** at unseen speeds (≥ 40 Hz)
- **0.747** at the largest unseen settings

Development folds gave 0.983 to 0.994.

Q3 Is it unfamiliar?

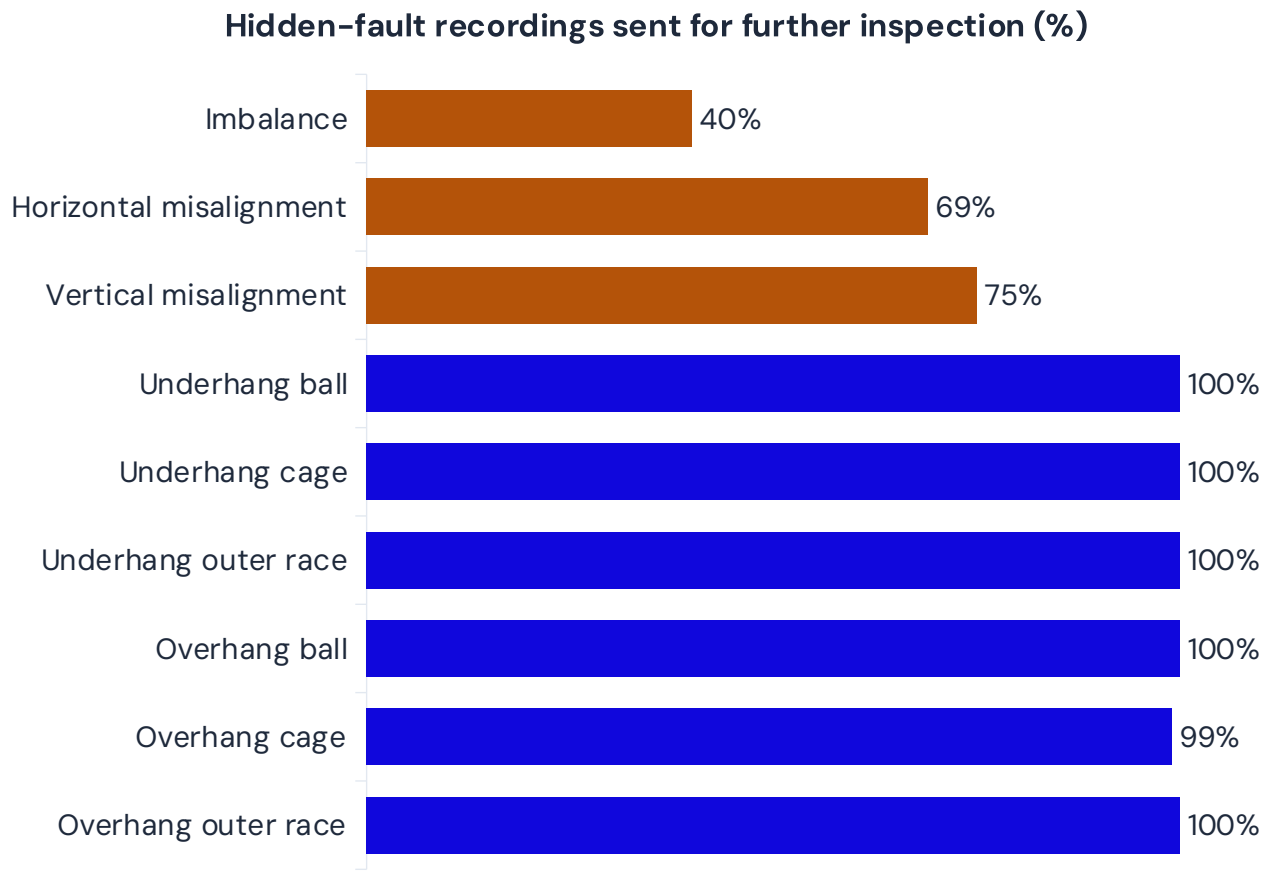
82.4 %

of unfamiliar faults sent for inspection

- **925 / 1,123** hidden-fault recordings
- **0.879** macro AUROC (Mahalanobis)
- **32.9 %** of known-side scorings also rejected

Rejection rose at unseen speeds and settings.

Unfamiliar faults: bearings caught, shaft faults harder



Bearing faults

99 to 100 % sent for inspection: no known class looks like them.

Imbalance and vertical misalignment

Often given another known shaft-fault label (114 and 44 recordings).

Horizontal misalignment

19 recordings called healthy: the most serious miss.

Why

In the CNN-6 feature space, an unseen shaft fault overlaps known shaft faults, so distance alone struggles.

Evaluating the investigation assistant

D7 pilot

225 answers: 15 cases × 5 questions × 3 arms, blind-scored

0 unsupported repair recommendations, in any arm

31 % unnecessary refusals with the code review on (1 % without)

15 / 15 correct sensor named with the review on

App-9: conversation mode

72 answers: 6 scenarios × 6 questions, two modes

69.4 % directly responsive (single question: 38.9 %)

8.6 % provenance errors in reviewed drafts (was 25.9 %)

0 cross-case leaks and 0 repair advice shown

Development cases, small samples, author-scored: read these as pilot results, not proof of safety.

Challenges along the way

99.3 %

A shortcut in the microphone

Microphone bursts in 168 files were linked to two classes: a simple classifier on them reached 99.3 % precision. The audit found them before any training.

1.000

A perfect score that was wrong

Random window splits scored 1.000 because pieces of one recording sat on both sides. Folder-grouped folds gave 0.878.

78 %

Healthy rejections follow speed

Healthy recordings below the calibrated speed range were not called plain healthy in 78 % of runs. The gap was data coverage, not the model.

1

improvement cleared its bar

Of the research stages declared in advance, only E1 (specialist autoencoders) passed: +10.1 points unknown recall on 3 seeds. Not deployed: found after the freeze.

Limitations and what comes next

Limitations

- One rig with induced faults: no claim about other machines
- 49 healthy recordings from one folder: healthy false-alarm rates are uncertain
- Folder grouping is only a proxy for recording sessions
- The sealed tests are now spent: new ideas need new held-out data
- The assistant pilot is small and scored by its author

What comes next

- Test E1 on a new held-out set
- Collect healthy recordings across speeds and sessions
- Evaluate the fine-tuned ResNet-18 on all nine hidden faults
- Relax over-strict assistant checks; use an independent scorer
- Test on a second rig before claiming transfer

Say what the evidence supports, and no more

Three questions from one recording, 0.916 macro-F1 on sealed tests, 82 % of unfamiliar faults sent for inspection, and every verdict shown with its evidence and its limits.

github.com/bhuveshsharma09/MachineSense

Bhuvesh Kumar · NUS-ISS